

§5.2

Mean and Standard Deviation of a Discrete Distribution

Calculating and interpreting the expected value of a random variable

Today's Goal

After completing this section, you should be able to:

- Calculate the mean (expected value) of a discrete probability distribution
- Calculate the variance and standard deviation of a discrete probability distribution
- Interpret the expected value in real-world contexts
- Apply expected value to decision-making problems

Learning Target 8

LT.8 Discrete Probability Distributions

I understand what a discrete probability distribution is (including binomial distributions). I can represent the discrete probability distribution of a given random variable by creating an appropriate table or graph and can use it to calculate the probability of an event. I can also calculate the mean, variance, and standard deviation for a discrete probability distribution.

A Free Throw Bet



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Suppose Steph Curry offers you the following bet:
“If I make my next free throw, you owe me \$100. If I miss
it, I owe you \$500.”

Steph Curry’s career free throw percentage is 91%.

Would you take this bet?

Population Parameters

Since a probability distribution describes all possible values of a random variable and their probabilities, it represents an entire **population** rather than a sample.

Therefore, when we calculate the mean and standard deviation, these are **parameters** (denoted by μ and σ), not statistics.

- Mean: μ , not $\bar{x}$
- Standard deviation: σ , not s
- Variance: σ^2 , not s^2

Mean (Expected Value)

Theorem 5.2.1 Mean of a Discrete Probability Distribution

The **mean** of a discrete probability distribution is given by

$$\mu = \sum [x \cdot P(x)].$$

This value is also called the **expected value** of the random variable, often denoted $E(X)$ or simply E .

The expected value represents the long-run average: if we repeated the experiment many times and recorded the value of X each time, the average of all those values would approach μ .

The Mean for Three Flips

A fair coin is flipped three times. Let X represent the number of heads obtained. Find the mean of this probability distribution.

x	$P(x)$	$x \cdot P(x)$
0	0.125	$0 \cdot 0.125 = 0.000$
1	0.375	$1 \cdot 0.375 = 0.375$
2	0.375	$2 \cdot 0.375 = 0.750$
3	0.125	$3 \cdot 0.125 = 0.375$
Total		$\mu = 1.500$

The mean is $\mu = 1.5$. If we flip a fair coin three times, we expect to get 1.5 heads on average.

Variance and Standard Deviation

Theorem 5.2.3 Variance and Standard Deviation of a Discrete Distribution

The **variance** of a discrete probability distribution is

$$\sigma^2 = \sum [(x - \mu)^2 \cdot P(x)].$$

The **standard deviation** is the square root of the variance:

$$\sigma = \sqrt{\sum [(x - \mu)^2 \cdot P(x)]}.$$

The Variance for Three Flips

Find the variance and standard deviation for the coin flip distribution from Example 1. Round to 2 decimal places.

We already found $\mu = 1.5$. Now we add a column for $(x - \mu)^2 \cdot P(x)$:

x	$P(x)$	$x \cdot P(x)$	$(x - \mu)^2 \cdot P(x)$
0	0.125	0.000	$(0 - 1.5)^2 \cdot 0.125 = 0.28125$
1	0.375	0.375	$(1 - 1.5)^2 \cdot 0.375 = 0.09375$
2	0.375	0.750	$(2 - 1.5)^2 \cdot 0.375 = 0.09375$
3	0.125	0.375	$(3 - 1.5)^2 \cdot 0.125 = 0.28125$
Total		$\mu = 1.500$	$\sigma^2 = 0.75$

The variance is $\sigma^2 = 0.75$ and the standard deviation is $\sigma = \sqrt{0.75} \approx 0.87$.

Now You Try: Mean and Variance

You Try 5.2.1. A river drawbridge has to be raised for anything too tall to pass beneath it, and the bridge house writes down every raising in the day's log. One day is chosen at random. Let X be the number of times the bridge is raised on that day.

x	0	1	2	3
$P(x)$	0.10	0.30	0.40	0.20

Round the mean, the variance and the standard deviation to **two** decimal places.

(a) Find the mean of this probability distribution.

(b) Find the variance.

(c) Find the standard deviation.

Now You Try: Mean and Variance

You Try 5.2.1, continued.

(d) Which statement says what the mean tells you here?

- The next day will have this many bridge raisings.
- This is the number of bridge raisings that occurs most often.
- Over a great many days, the average number of bridge raisings per day would be close to this value.
- Half of all days have fewer bridge raisings than this, and half have more.

You Try 5.2.1: Checking Your Answer

(a) Mean

$$\mu = 0(0.10) + 1(0.30) + 2(0.40) + 3(0.20) = 1.70$$

(b) Variance

$$\sigma^2 = 0.289 + 0.147 + 0.036 + 0.338 = 0.81$$

(c) Standard deviation

$$\sigma = \sqrt{0.81} = 0.90$$

(d) What the mean tells you

Over a great many days, the average number of bridge raisings per day would be close to this value.

The Free Throw Bet

Suppose Steph Curry offers you the following bet: “If I make my next free throw, you owe me \$100. If I miss it, I owe you \$500.” Steph Curry’s career free throw percentage is 91%. What is the expected value of this bet for you?

Let X represent your net gain from the bet. We can set up a table.

Event	x	$P(x)$	$x \cdot P(x)$
Curry makes it	−\$100	0.91	−\$91
Curry misses	+\$500	0.09	+\$45
Total			$E(X) = -\$46$

The expected value is −\$46. This means that if you took this bet many times, you would lose an average of \$46 per bet. This is not a good bet for you.

A Raffle Ticket

A service organization sells 500 raffle tickets for \$2 each. First prize is \$250, second prize is \$100, and third prize is \$50. Each ticket has an equal chance of winning.

- (a) Construct the probability distribution for X = net gain from purchasing one ticket.
- (b) Find the probability of winning any prize.
- (c) Find the expected value of X . Round to 2 decimal places.

The Net Gain From One Ticket

A service organization sells 500 raffle tickets for \$2 each. First prize is \$250, second prize is \$100, and third prize is \$50.

(a) Your net gain is the prize minus the \$2 cost of the ticket.

Event	x (Net Gain)	$P(x)$
Win 1st Prize	$\$250 - \$2 = \$248$	$\frac{1}{500} = 0.002$
Win 2nd Prize	$\$100 - \$2 = \$98$	$\frac{1}{500} = 0.002$
Win 3rd Prize	$\$50 - \$2 = \$48$	$\frac{1}{500} = 0.002$
Win Nothing	$\$0 - \$2 = -\$2$	$\frac{497}{500} = 0.994$

(b) $P(\text{winning}) = 0.002 + 0.002 + 0.002 = 0.006 = 0.6\%$.

The Expected Value of a Raffle Ticket

(c) Find the expected value of X . Round to 2 decimal places.

Event	x	$P(x)$	$x \cdot P(x)$
Win 1st Prize	\$248	0.002	\$0.496
Win 2nd Prize	\$98	0.002	\$0.196
Win 3rd Prize	\$48	0.002	\$0.096
Win Nothing	−\$2	0.994	−\$1.988
Total			$E(X) = -\$1.20$

The expected value is −\$1.20. On average, you lose \$1.20 for each raffle ticket purchased. However, the organization raises money for a good cause, and many people enjoy the excitement of possibly winning.

Now You Try: Expected Value

You Try 5.2.2. A student buys a mystery box of returned items for \$20 and resells what is inside. Let X be the net gain from one box, in dollars.

(a) Complete the table to find the expected net gain.

Sells for	Net gain x	$P(x)$	$x \cdot P(x)$
\$0	?	0.20	?
\$20	?	0.45	?
\$50	?	0.35	?
Total			?

Now You Try: Expected Value

You Try 5.2.2, continued.

- (b) Over a great many boxes, does the student gain or lose on average?
- (c) What would the student have to pay for a box for the buy to be fair?
- (d) The store also offers a checked box at the same \$20. It is certain to sell for \$25. Which is the better buy in the long run?

You Try 5.2.2: Checking Your Answer

(a) Each box costs \$20, so the net gain on a row is what it sells for minus \$20.

Sells for	Net gain x	$P(x)$	$x \cdot P(x)$
\$0	−\$20.00	0.20	−\$4.00
\$20	\$0.00	0.45	\$0.00
\$50	\$30.00	0.35	\$10.50
Total			\$6.50

So the expected net gain is \$6.50.

You Try 5.2.2: Checking Your Answer

(b) Over a great many boxes, does the student gain or lose on average?

The student gains on average.

(c) What would the student have to pay for a box for the buy to be fair?

$$\$0(0.20) + \$20(0.45) + \$50(0.35) = \$26.50$$

(d) Which is the better buy in the long run?

The mystery box. Over a great many purchases it returns more on average.

The checked box is certain to sell for \$25, so its net gain is $\$25 - \$20 = \$5.00$.

The expected net gain of the mystery box is \$6.50.

Interpretation of Expected Value

The expected value has important interpretations.

1. **Long-run average.** If the random experiment is repeated many times, the average of all outcomes will approach the expected value.
2. **Fair value.** A game or bet is considered **fair** if the expected value is zero—neither player has an advantage in the long run.
3. **Decision criterion.** When comparing options with uncertain outcomes, the option with the higher expected value is often (but not always) preferred.

Expected Value and a Single Trial

Warning 5.2.7

Expected value represents an average over many repetitions. It does *not* predict what will happen on any single trial. For example, if the expected value of a game is $-\$0.40$, you will never actually lose exactly 40 cents—you either win a prize or lose $\$1$. The expected value only tells you the average outcome over many plays.

An Insurance Policy

An insurance company sells a 1-year policy for \$500. If the insured person dies during the year (probability 0.002), the company pays \$200,000.

- (a) Find the company's expected profit per policy.

$$\begin{aligned} E(X) &= \$500(0.998) + (\$500 - \$200,000)(0.002) \\ &= \$499 - \$399 = \$100 \text{ per policy} \end{aligned}$$

- (b) If they sell 10,000 policies, what is their expected total profit?

$$10,000 \times \$100 = \$1,000,000$$